

1. WAP to print whether a given number is even or odd.
2. WAP to read name, gender, acc_number from the user and print them.
3. WAP to check if the number is positive or negative.
4. WAP to swap two numbers.
5. WAP to swap two numbers without using third variable.
6. WAP to check if number is divisible by 5 or not.
7. WAP to check if number is divisible by 3 and 7.
8. WAP to print whether a given number is pos-even / pos-odd or neg-even / neg-odd or zero.
9. WAP to print greatest of two numbers.
10. WAP to print greatest of three numbers.
11. WAP to print smallest of two numbers.
12. WAP to find the middle number.
13. WAP to check if the sides of triangle form right angle triangle.
14. WAP to implement calculator performing addition (+), subtraction (-), multiplication (*), division (/).
15. WAP shows bank details of total balance and allows user to choose either to continue or exit and withdrawal or deposit.
If withdrawal it checks the balance and shows a message insufficient balance and for deposit it adds up to total balance and repeats until user choose to exit.
16. WAP to determine grade based on the given condition:
Marks : 80- 100 : A
Marks : 65 – 79 : B
Marks : 50 - 64 : C
Marks : 35 – 49 : D
Marks < 35 : Fail
17. WAP to determine whether given character is a digit, lower case, upper case, or a special character.
18. WAP to print the direction based on the first letter using switch case.
19. WAP to print whether the character is vowel or not by using switch.
20. WAP to print square or cube of given number based on condition using switch case.
21. WAP to implement calculator performing addition (+), subtraction (-), multiplication (*), division (/) using switch case.
22. WAP to calculate the ticket price based on the age
Age : under 5 – free
Age : 5-12 :- 50 % discount
Age : 65 and above :- 20 % discount
Other age groups regular price.
23. WAP to print greeting based on time
Good morning! - 5-11
Good afternoon! - 12-15
Good evening! -16-21
Good night! - 22-4
Invalid hour >24
24. WAP to find the minimum number of notes required for a given amount.
25. WAP to check whether the given number is prime or not.
26. WAP to print 'n' natural numbers.
27. WAP to print sum of 'n' natural numbers.
28. WAP to find factorial of a number.
29. WAP to find H.C.F and L.C.M of given two numbers.
30. WAP to print the multiples of 5 .
31. WAP to print the multiples of 5 without using *.
32. WAP to find the largest of N numbers with loops.

33. WAP to find the number of digits in a given number 'n'.
34. WAP to print the reverse of number "n'.
35. WAP to check whether a number is palindrome or not.
36. WAP to check if the number is Armstrong number or not.
37. WAP to print n prime numbers from 2 to n.
38. WAP to print n prime numbers in given range.
39. WAP to print n palindrome numbers.
40. WAP to check if the number is perfect number or not.
41. WAP to print n perfect numbers.
42. WAP to check if given year is leap year or not.
43. WAP to count the number of prime numbers in the given range.
44. WAP to read the binary and convert into decimal.
45. WAP to print binary value of a number.
46. WAP to generate n Fibonacci numbers.
47. WAP to generate n negative Fibonacci numbers.
48. WAP to replace 0's with 1's in a given number.
49. WAP to print the next prime number.
50. WAP to swap the first and last digits of a given number.
51. WAP to print greatest of two numbers using ternary operator.
52. WAP to print greatest of three numbers using ternary operator.
53. WAP to read and print values in a n array.
54. WAP to reverse and print the array elements without using another array.
55. WAP to store the square of each element in an array.
56. WAP to check the given array is palindrome or not.
57. WAP to print the frequency of elements in an array.
58. WAP to sum the array values in even index.
59. WAP to check if two arrays are equal or not.
60. WAP to read the elements of an array and check whether search element is present or not.
61. WAP to check if the subarray is present or not.
62. WAP to sort the array in ascending order.
63. WAP to sort the array in descending order.
64. WAP to find the sum of the digits of a number till it becomes to a single digit. (eg: 178->16 -> 7)
65. WAP to print the following series: 16 11 6 1 6 11 16
66. WAP to make the user guess a secret number.
There will be a 'n' chances to find it. It is the user's choice either to use hint or not.
If no hint 'n' chances else 'n-2' chances.
67. WAF to print next number.
68. WAF to print if a number is even or odd.
69. WAF to print the factorial of a number.
70. WAF to print if a number is palindrome or not.
71. WAF to add digits of a number
72. WAF to print all prime number upto limit.

```

*
* *
* * *
* * * *
* * * * *

```

Right Half Pyramid

```

                *
            * *
        * * *
    * * * *
* * * * *

```

Left Half Pyramid

```

      *
    * *
  * * *
* * * *
* * * * *

```

Full Pyramid

```

* * * * *
* * * *
* * *
* *
*

```

Inverted Right Half Pyramid

```

* * * * *
  * * * *
    * * *
      * *
        *

```

Inverted Left Half Pyramid

```

* * * * *
  * * * *
    * * *
      * *
        *

```

Inverted Full Pyramid

```

* * * *
* * * *
* * * *
* * * *
* * * *

```

Rhombus Pattern

```

      *
    * *
  * * *
* * * *
* * * *
  * * *
    * *
      *

```

Diamond Pattern

```

*   *   *   *
 *  *  *  *
  * * * *
   *
  * * *
 *  *  *
*   *   *   *

```

Hourglass Pattern

```

* * * * *
*       *
*       *
*       *
*       *
*       *
*       *
* * * * *

```

Hollow Square Pattern

```

      *
    * *
  *   *
 *     *
*       *
* * * * *

```

Hollow Full Pyramid

```

* * * * *
 *       *
  *     *
   *   *
    * *
     *

```

Hollow Inverted Full Pyramid

```

      *
    * *
  * * *
* * * *
* * * *
  * * *
    * *
      *

```

Hollow Diamond Pyramid

```

* * * * *
 *  *  *
  * * *
   *
  * * *
 *  *  *
* * * * *

```

Hollow Hourglass Pattern

```

1
2 3
4 5 6
7 8 9 10

```

Floyd's Triangle

```

      1
    1 1
  1 2 1
1 3 3 1

```

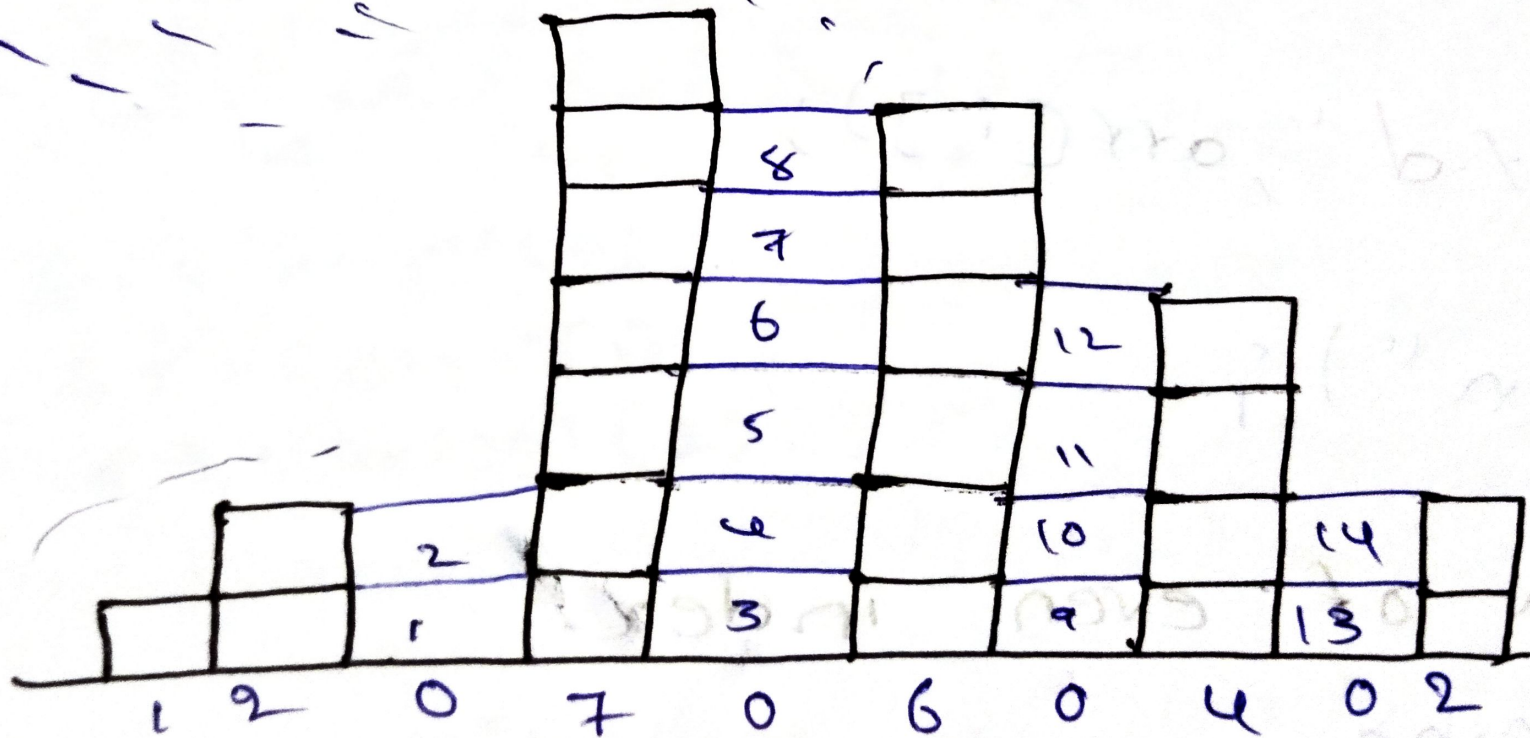
Pascal's Triangle

1	1	1	1	1	54321	12345
22	12	2 3	21	12	4321	1234
333	123	4 5 6	321	123	321	123
4444	1234	7 8 9 10	4321	1234	21	12
55555	12345	11 12 13 14 15	54321	12345	1	1

[illegible]

water trapping:-

```
int arr[] = { 1, 2, 0, 7, 0, 6, 0, 4, 0, 2 }
```



Basic Refresher

Easy :

If :

1. WAP to check numbers between 20 and 100.
2. WAP to check if a number is -ve or +ve or 0.
3. WAP to find the largest of two numbers.
4. WAP to check whether a character is Upper case Alphabet.
5. WAP to check whether a character is digit or not.
6. WAP to check whether a character is Lowercase Alphabet.
7. WAP to print the result as Pass or Fail based on condition
Marks > 35 : Pass
Marks <= 35 : Fail
8. WAP to check an integer number is even or odd.
9. WAP to check if a person is eligible to vote or not (voting age >= 18).
10. WAP to check a number is divisible by both 2 and 3.

Switch :

1. Based on choice of user you should find the square and cube of a given number.
2. Check if a given character is a digit or lowercase/uppercase alphabet or none of these, using switch case.
3. Take two numbers from user and perform addition/subtraction Using a switch case.

Loops :

1. Patterns:

```
* * * * *      *      * * * * *      * * * * *      *
* * * * *      * *      * * * *      * * * *      * *
* * * * *      * * *      * * *      * * *      * * *
* * * * *      * * * *      * *      * *      * * * *
* * * * *      * * * * *      *      *      * * * * *
```

* * * * *	1 1 1 1 1	* *	1
*	2 2 2 2 2	* *	1 2
* * * * *	3 3 3 3 3	*	1 2 1
*	4 4 4 4 4	* *	1 2 1 2
* * * * *	5 5 5 5 5	* *	

Arrays:

1. WAP to read the integer array elements and print all the elements of an array.(Read the size of an Array from the user).
2. WAP to print even indices elements of an character Array.
3. WAP to print odd indices elements of an float Array.

Bitwise Operator:

- 1.WAP to check if the number is even or odd.
2. WAP to set,clear,toggle the nth bit from a given number..
- 3.WAP to print bits of a number.
- 4.WAP to count set bits of number.

Medium :

If :

1. WAP to find the largest among three numbers.
2. WAP to find the smallest among three numbers.
3. WAP to find the middle (by value) number among three numbers.
4. WAP to check if a given year is a leap or not.
5. WAP to print the grade A , B, C, D, E and F as given below:

marks(91 - 100) : A
marks(81 - 90) : B
marks(71 - 80) : C
marks(61 - 70) : D
marks(51 - 60) : E
marks(<= 50) : F

Switch :

1. WAP to perform basic calculator using switch case.
2. WAP to print the grade for the marks using switch case.

Loops :

1. WAP to check if a number is prime or not.
2. WAP to check if a number is perfect or not.
3. WAP to print 2 power series from 2^1 to limit 2^L .
4. WAP to print number of digits in number.
5. WAP to find the sum of all the digits of a positive integer number.
6. WAP to print sum of 1st N natural numbers.
7. WAP to find the factorial of a given number.
8. WAP to print A.P. series for given a, d, N, values,
where , a = starting value
d = common difference
N = Total elements in series
9. WAP to print X^Y value for X and Y given by the user.
10. WAP to print the reverse of the given number.
11. WAP to print positive fibonacci series up to the limit L.
12. WAP to print all prime numbers up to the limit L.
13. WAP to print negative fibonacci series up to the limit L.
14. WAP to print first N elements of positive fibonacci series.
15. Patterns:

1	1	1	#	#
1 2	1 0	2 3	#	#
1 2 3	1 0 1	4 5 6	@	
1 2 3 4	1 0 1 0	7 8 9 10	#	#
			#	#

1	1	1	1
2 2	2 2	0 1	2 3
3 3 3	1 1 1	1 0 1	4 5 6

4 4 4 4	2 2 2 2	0 1 0 1	7 8 9 10
5 5 5 5 5	1 1 1 1 1	1 0 1 0 1	

*****	*****	*	1
* *	*****	***	10
* *	***	*****	101
*****	*	*****	1010

A B C D E
F G H I J
K L M N O
P Q R S T
U V W X Y

Arrays:

1. WAP to calculate the average of all elements in an integer array.
2. WAP to find the largest element in an integer array.
3. WAP to find the smallest element in an integer array.
4. WAP to reverse the elements of an integer array. For example, if the array is [1, 2, 3, 4, 5], the reversed array should be [5, 4, 3, 2, 1].
5. WAP to sort the elements of an integer array in ascending order using the bubble sort algorithm.
6. WAP to copy the elements of one integer array into another array.

Bitwise operator:

1. WAP to set,toggle,clear n bits from the pos position of given number.
2. WAP to check number is power of 2 or not
3. WAP to check Mth bit of num is set or not,if yes toggle Nth bit ,else clear Nth bit and print the updated value num.

4. WAP to input any number from the user and find the total number of leading zeros of the given number. (i/p=22 o/p: Leading Zeros=27).
5. WAP to input any number from the user and find the total number of trailing zeros of the given number.(i/p=22 o/p: Trailing Zeros=1).

Hard :

If :

1. Write a C program to input all sides of a triangle and check whether triangle is valid or not.
2. Write a C program to input amount from user and print minimum number of notes (Rs. 500, 100, 50, 20, 10, 5, 2, required for the amount.
3. Write a C program to input electricity unit charge and calculate the total electricity bill according to the given condition:
For first 50 units Rs. 0.50/unit
For next 100 units Rs. 0.75/unit
For next 100 units Rs. 1.20/unit
For unit above 250 Rs. 1.50/unit
an additional surcharge of 20% is added to the bill.

Switch :

1. Write a C program to input two numbers from user and find maximum between two numbers using switch case.

Loops :

1. WAP to print 1st N prime numbers.
2. WAP to check if a number is prime or not. if prime print message Otherwise print all the divisors of that number.
3. WAP to check if a three digit number is armstrong or not.
4. WAP to scan n elements from user and tell the largest among them without using the array.
5. WAP to scan n elements from user and tell the smallest among them without using the array.
6. WAP to take n value from user, print natural number from 1 to n*10. Where you have to print 1st n numbers and skip another n numbers then print next n numbers then skip Next n numbers. Ex :

i/p : 3

o/p : 1 2 3 7 8 9 13 14 15 19 20 21 25 26 27.

Patterns:

```
1 10 11 20 21
2 9 12 19 22
3 8 13 18 23
4 7 14 17 24
5 6 15 16 25
```

```
1
2 6
3 7 10
4 8 11 13
5 9 12 14 15
```

```
1 2 3 4 5
10 9 8 7 6
11 12 13 14 15
20 19 18 17 16
21 22 23 24 25
```

```
    1
  1 0 1
1 0 1 0 1
1 0 1 0 1 0 1
```

```
9 8 7 6 5
4 3 2
  1
2 3 4
```

1 0 1 0 1 0 1 0 1

5 6 7 8 9

Arrays:

1. WAP to find the median of given Array.
2. WAP to print frequency (occurrence) of all elements of an Array.
3. WAP to remove duplicates in an array. Return a new array without duplicates and size of new array.
4. WAP to concatenate two integer arrays into a new array. For example, if the arrays are [1, 2, 3] and [4, 5, 6], the concatenated array should be [1, 2, 3, 4, 5, 6].
5. WAP to check if an integer array is a palindrome (reads the same forwards and backwards).
6. WAP that finds the intersection of two integer arrays, i.e., the elements that are common to both arrays. ([1,2,3] and [2,6,3], the common elements are 2 and 3)
7. WAP that finds the union of two integer arrays, i.e., combination of elements of both arrays without repetition. ([1,2,3] and [2,6,3], the union of these arrays: 1,2,3,6)

Bitwise operator:

- 1.WAP to swap two numbers using a bitwise operator.
- 2.WAP to rotate bits of number n times.Ask the user to rotate left or right side.

Functions and Pointers part 1

Easy :

1. WAF to find the factorial of a number.
2. WAF to find prime numbers within the range.
3. WAP to return an address from a function.
4. WAP using pointers
 - 1) Add two integers / floats.

- 2) Subtraction of two integers/floats.
- 3) Multiplication of two integers/floats.
- 4) Division of two integers / floats.

Medium :

1. WAF to swap nibbles using a function.
2. WAP to swap two integers without using temp Variable.
3. WAP to demonstrate Pass by value and pass by Reference.
4. WAP to swap two values using pass by reference Method without using temp variable.
5. WAP to fetch data from memory address 10000.

Hard :

1. Write a C program to check whether a number is prime, armstrong, perfect number or not using functions.

Recursion

Easy :

1. Write a C program to find the sum of elements in an array using a recursion.

Medium :

1. WAP to find the factorial of a number using recursion.
2. Write a C program to find the sum of digits of a number using recursion.

Hard :

1. Write a C program to find LCM of two numbers using recursion.

Passing Arrays to Functions

Easy :

1. Write a C program to pass an array to a function and display the elements of the array using that function.

Medium :

1. Write a C program to reverse the elements of an array by passing the array to a separate function.
2. Write a C program that copies the contents of one array to another by passing both arrays to a function.

Hard :

1. Write a C program to reverse the elements of an array by passing the array to a separate function.
2. Create a C program that takes an array as a parameter and replaces each element with 1 if it's a prime number and 0 if it's not, using a separate function.

STD IO

Easy :

1. Write a program that reads a sentence from the user and counts the number of words in the sentence.
2. Write a program that reads a sentence from the user and counts the number of vowels present in the sentence.

Medium :

Hard :

Strings

Easy :

1. WAP to compare the strings.
2. WAP to reverse a string.
3. WAP to string palindrome.
4. WAP to find string length / in a single line.

Medium :

1. WAP to reverse a string without string length.
2. WAP to convert the first and last letter of each word to capital in a line.
3. WAP to check unique strings.

Hard :

1. WAP to count the number of occurrences of a character in a string.
2. WAP to find the string is anagram or not.

3. Delete duplicates in character array without creating a new array

Storage Classes

Easy :

1. When the memory will be allocated for static const Int x = 5.
2. WAP for demonstrate static local variables.
3. WAP using extern storage class.
4. WAP to find factorial using void data type.

Medium :

Hard :